

The Geography of Income Inequality

EC3335 Set 12

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Content

(i) The rise of "big data"

(ii) "Big data" and the American Dream

(iii) Causal Effect of Neighborhoods

Housekeeping

PS03 (*due Friday, Nov. 25*)

- Posted now

PS04 (*due Sun, Dec. 4*)

TotC book report (*due Thu, Dec. 8*)

- Short (1000 words) assignment
- rubric posted on canvas and on the course repo
- will be checking for plagiarism

Final (*14:45 Wed, Dec. 7*)

- [final schedule](#)
- Comprehensive exam with an emphasis on the newer material

The rise of "big data"

Theoretical social science

Until very recently, social scientists were very limited by data to study these types of **policy questions**

Social science has been mostly a **theoretical field** to make policy recommendations

- Develop mathematical models (economics)
- Qualitative theories (sociology)

Problem: We were never able to test most of these theories

- Led to politicization of scientific questions such as

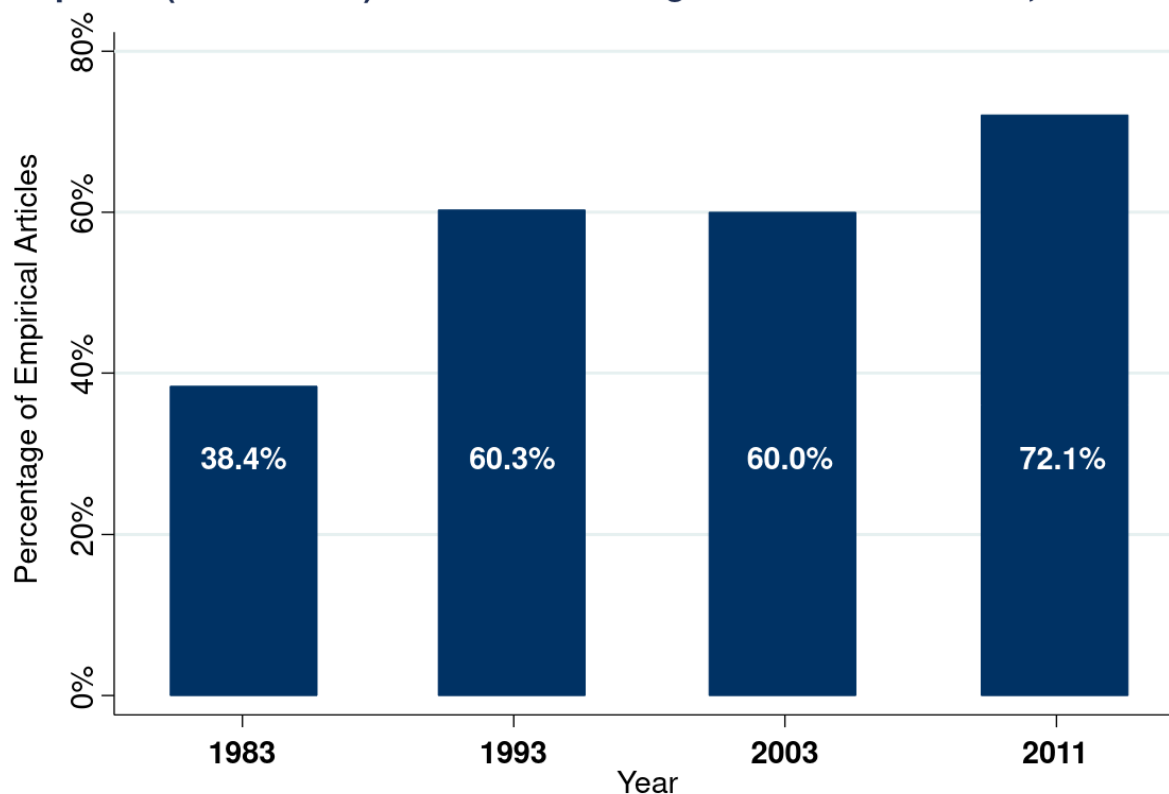
| Is Obamacare reducing job growth in America?

Big data and empirics

Social sciences are more empirical thanks to growing availability of data

- Test and improve theories using real world data

Empirical (Data-Based) Articles in Leading Economics Journals, 1983-2011



Source: Hamermesh (JEL 2013)

Big data and empirics

What is "**Big data**"?

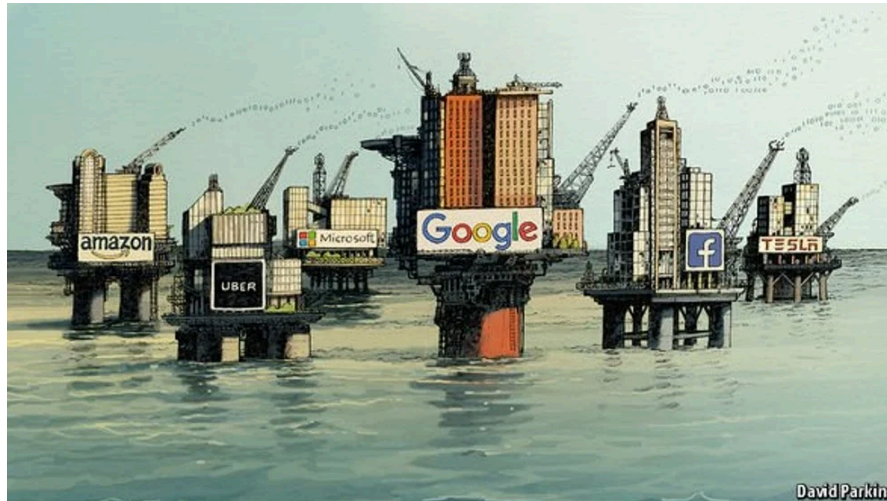
Large datasets

Where did "**Big data**" come from?

Product of the **internet age**

- Information costs $\rightarrow 0$

Product of post-internet private sector: "**Data is the new oil**"[†]



[†] Whatsapp, Robinhood

Big data and empirics

Examples:

- Government data: Tax records
- Corporate data: Google etc.
- Text data: Twitter, newspapers
- Satellite imagery
- Cellphone location data
- GitHub data

Benefits:

- More reliable than surveys
- New variables (eg emotions, locations)
- Rich coverage $\Rightarrow$ study subgroups
- Large samples

The American Dream †

† This lecture uses content from Raj Chetty's course: <https://opportunityinsights.org/course/>

The American Dream

The **American Dream** is an multi-faceted, abstract concept that may be interpreted in many ways

*Is the American Dream alive and well today? How should we **judge** that?*

How can we distill the concept of an the American Dream into a **statistic** that we can **systematically measure** using big data?

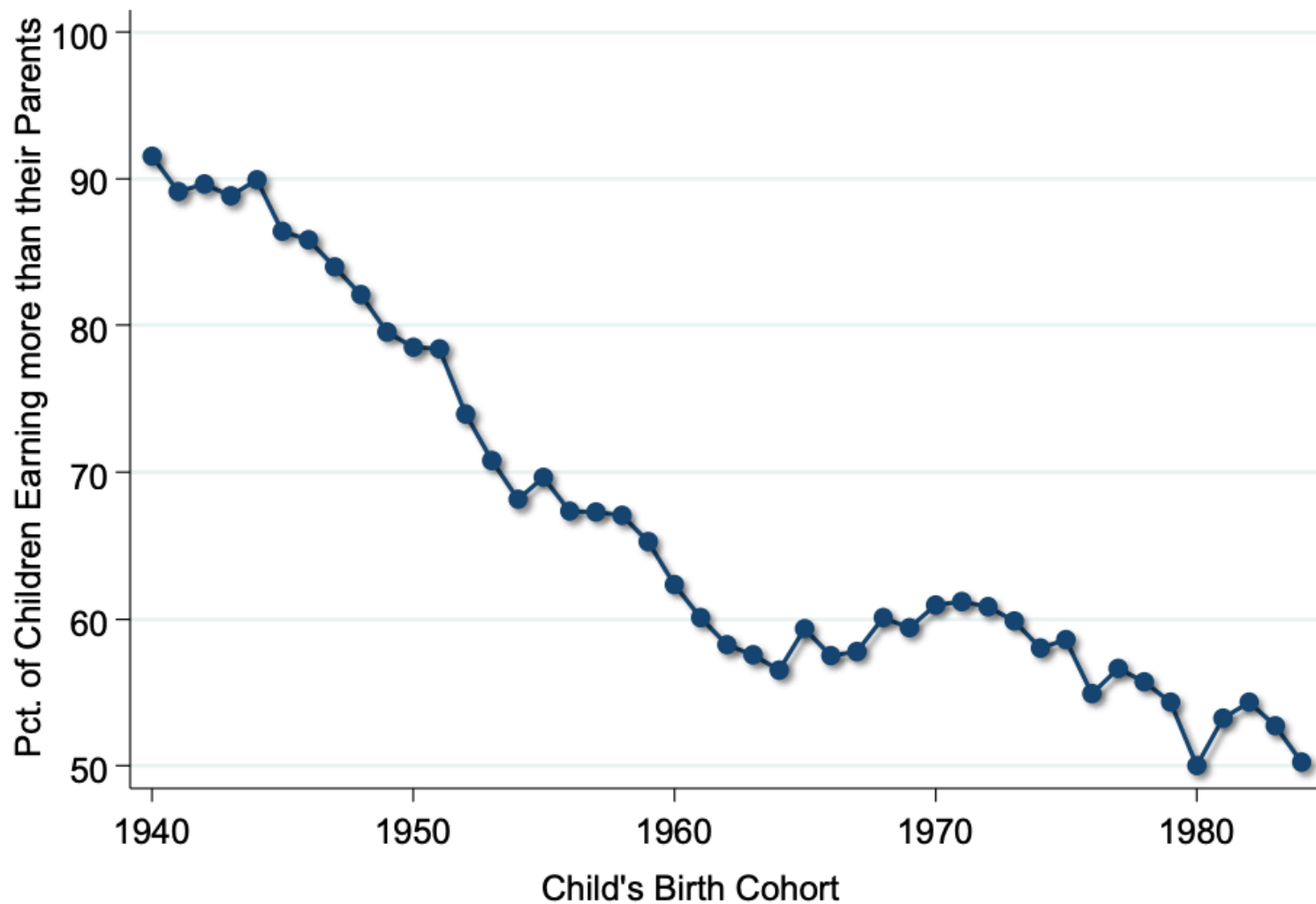
Obama (2014):

People's frustrations are partly rooted "in the fear that their kids won't be better off than they were."

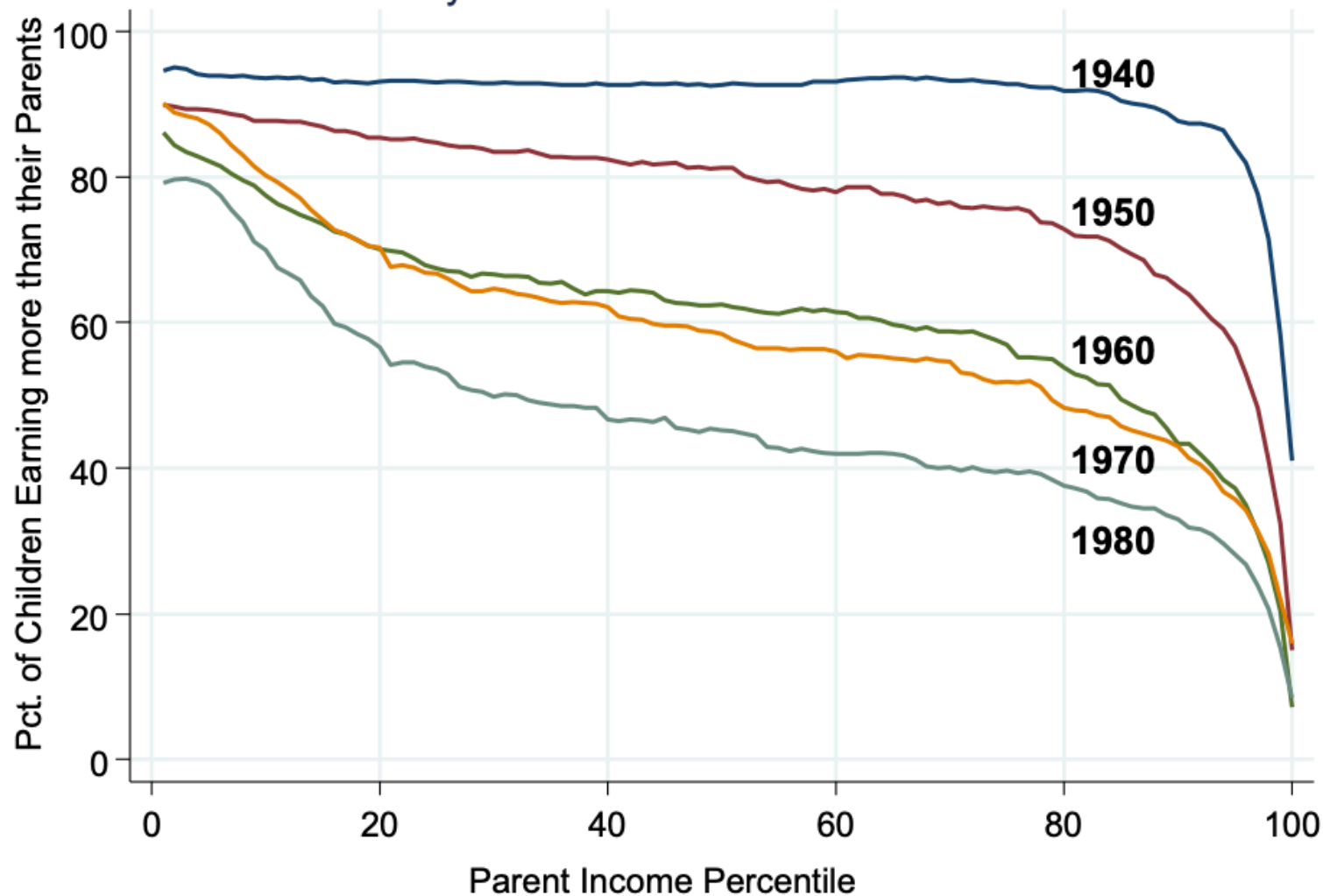
One **systematic measure** the American Dream: *What fraction of children earn more income than their parents? Is this a **fair** measure?*

We'd like to know: How has this statistic changed over time? Up or down?

Percent of Children Earning More than Their Parents, by Birth Cohort

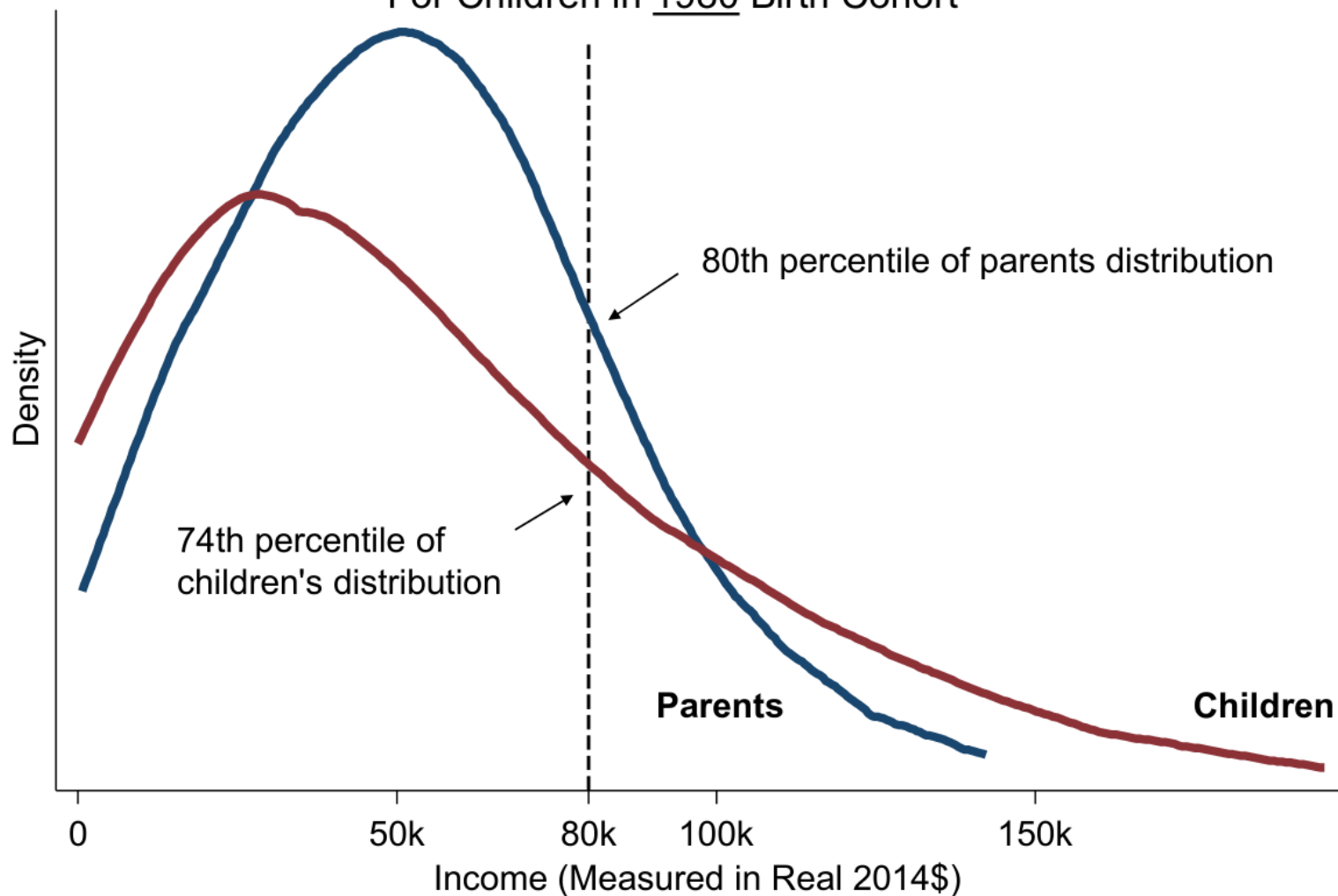


Percent of Children Earning More than their Parents By Parent Income Percentile



Household Income Distributions of Parents and Children at Age 30

For Children in 1980 Birth Cohort



The fading American Dream

What do these figures indicate about the American Dream?

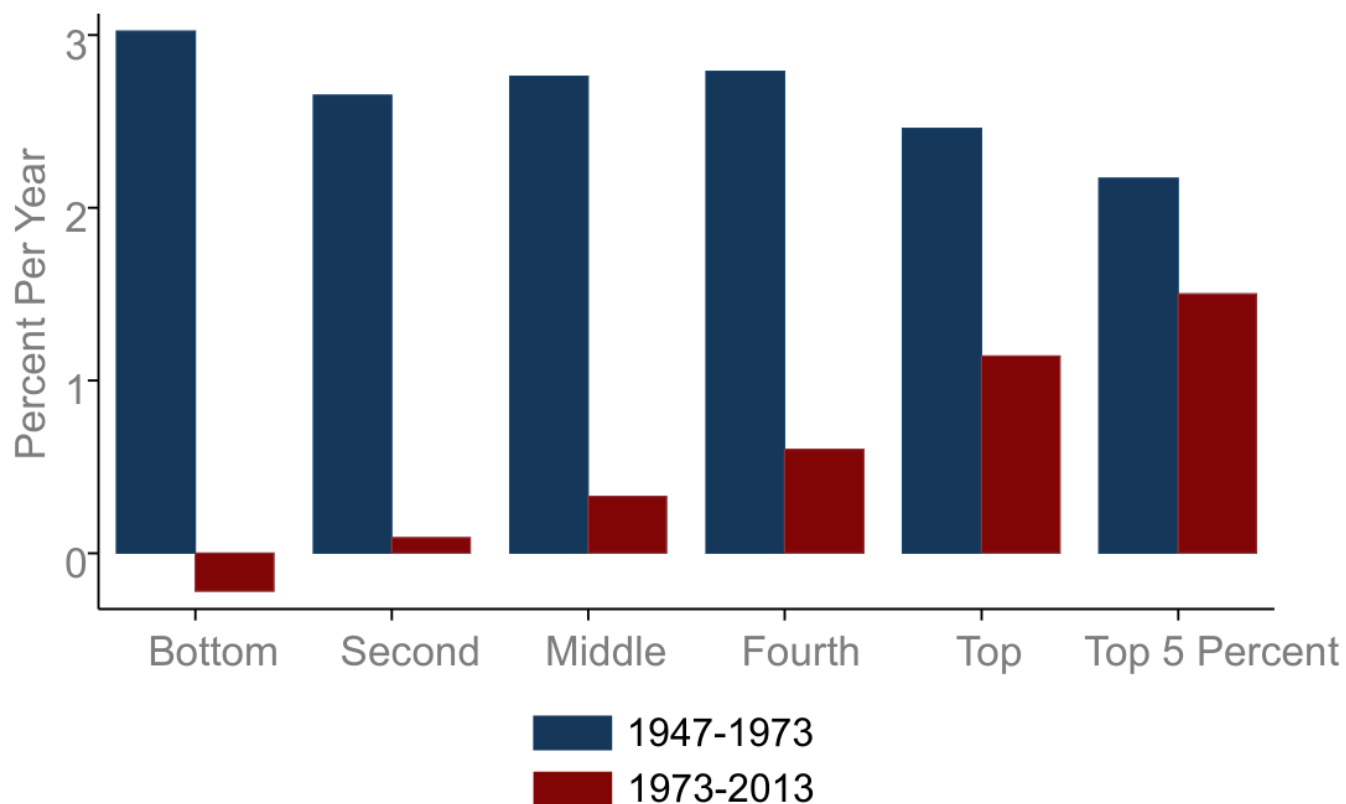
It has become harder and harder to reach it

Two major changes in the American economy since the 1940s:

- (i)** Lower total economic growth
- (ii)** Less equal distribution of growth

Growth in family income

The following chart compares growth in real mean family income by quintile between 1947-1973 and 1973-2013

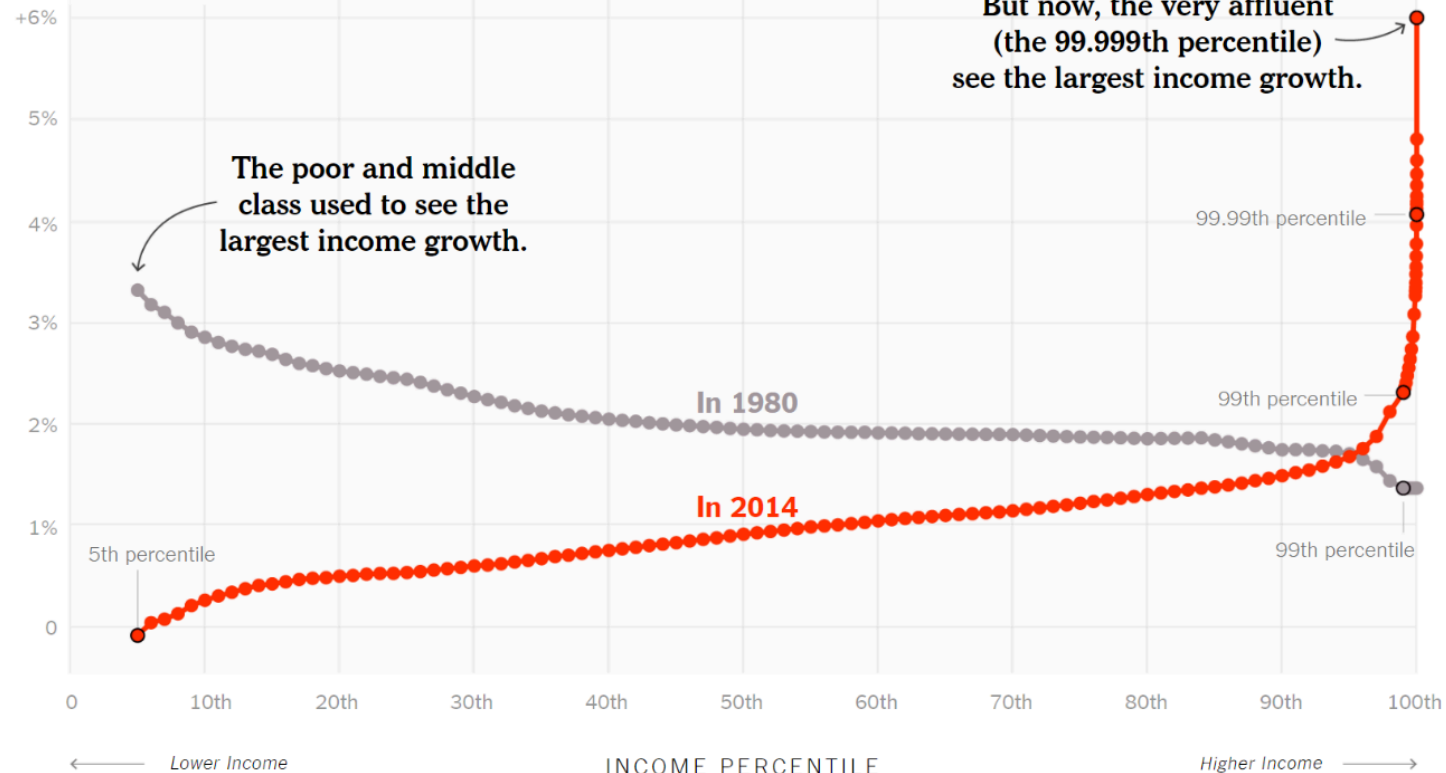


Source: Goldin and Katz (2007)

Growth in family income

INCOME GROWTH

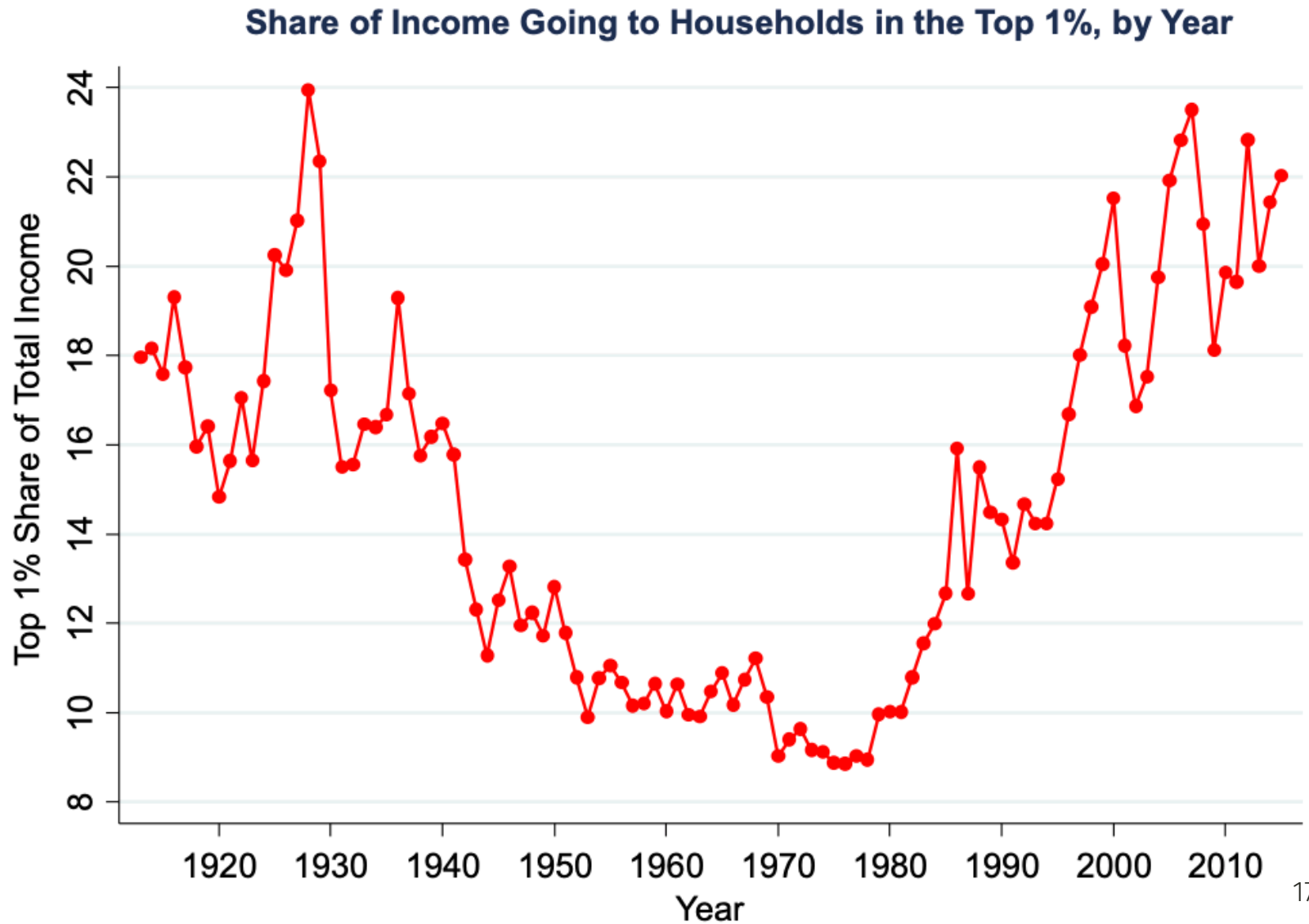
Over previous 34 years



Note: Inflation-adjusted annual average growth using income after taxes, transfers and non-cash benefits.

Source: Piketty, Saez, and Zucman (2017); Leonhardt (2017)

Growth in family income



Growth in family income

Two major changes in the American economy since the 1940s:

- Lower total economic growth
- Less equal distribution of growth

What policies can increase absolute mobility?

Consider two hypothetical scenarios for children born in 1980:

(i) Higher growth

1940 cohort growth rate with the 1980 cohort historical distribution

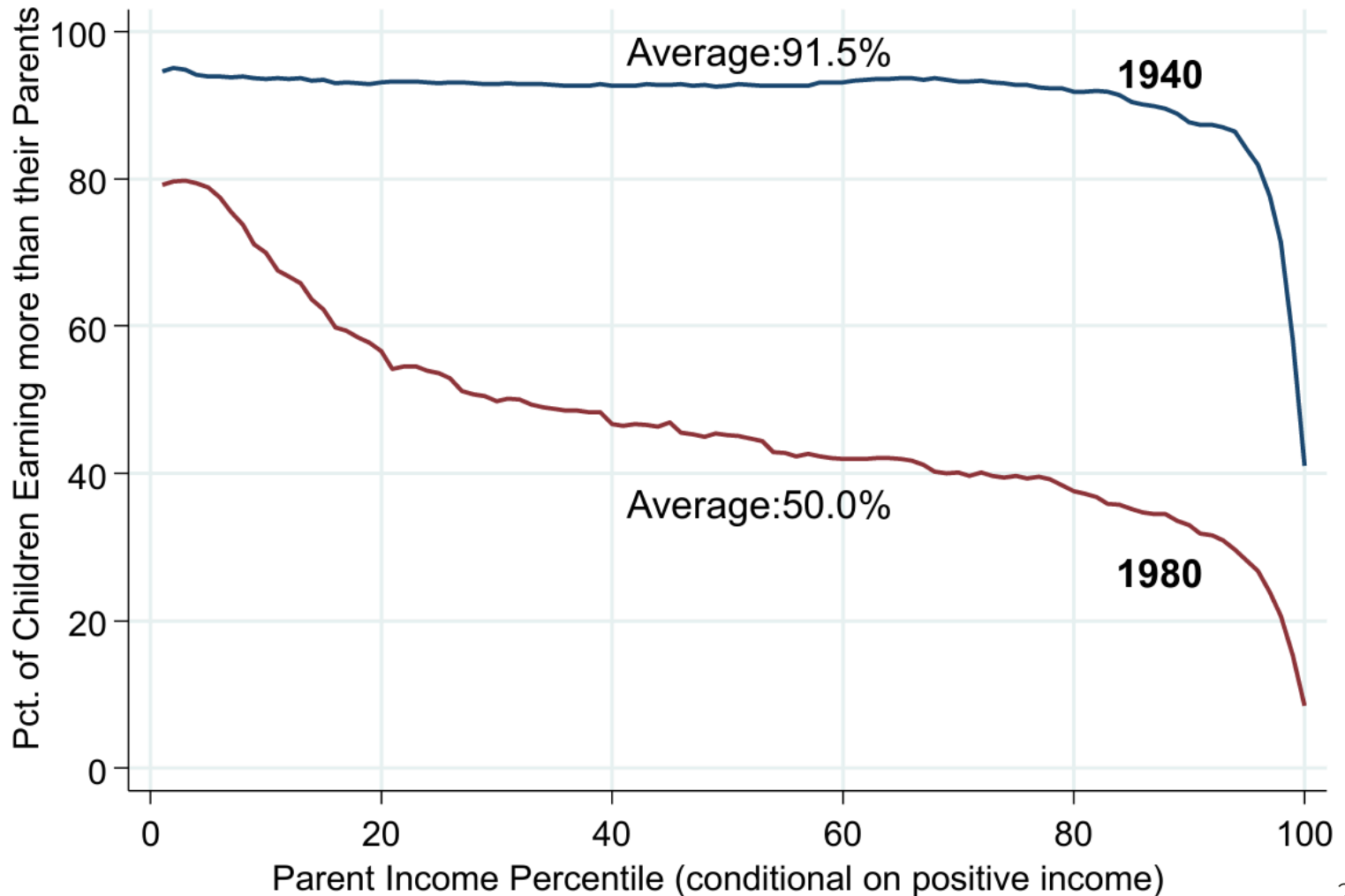
(ii) More broadly shared growth

Same 1980 cohort historical growth rates but 1940 cohort distribution across income groups

Which over these scenarios would have lead to a stronger American Dream?

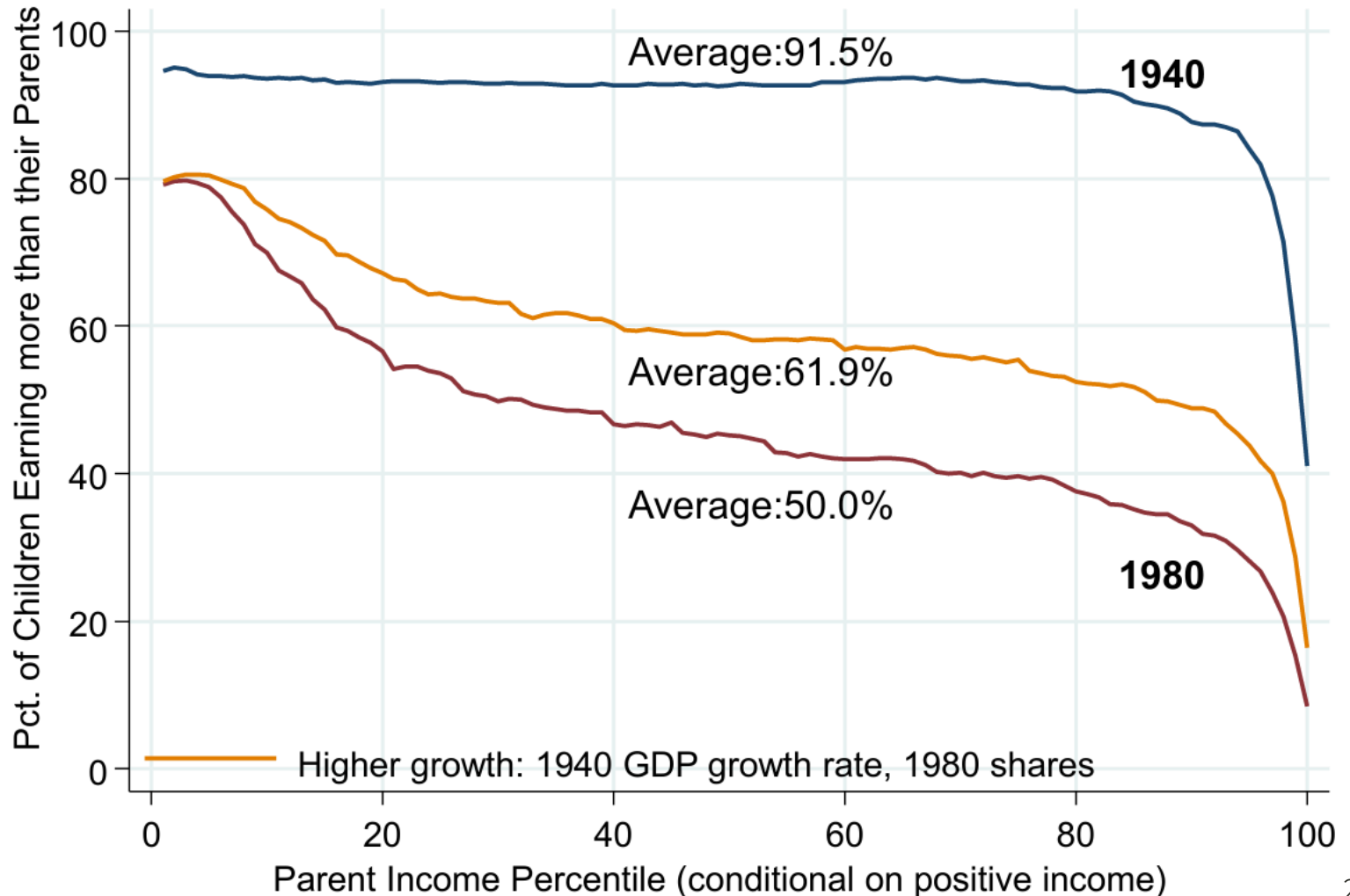
Hypothetical scenarios

Percent of Children Earning More than Their Parents: Hypothetical Scenarios



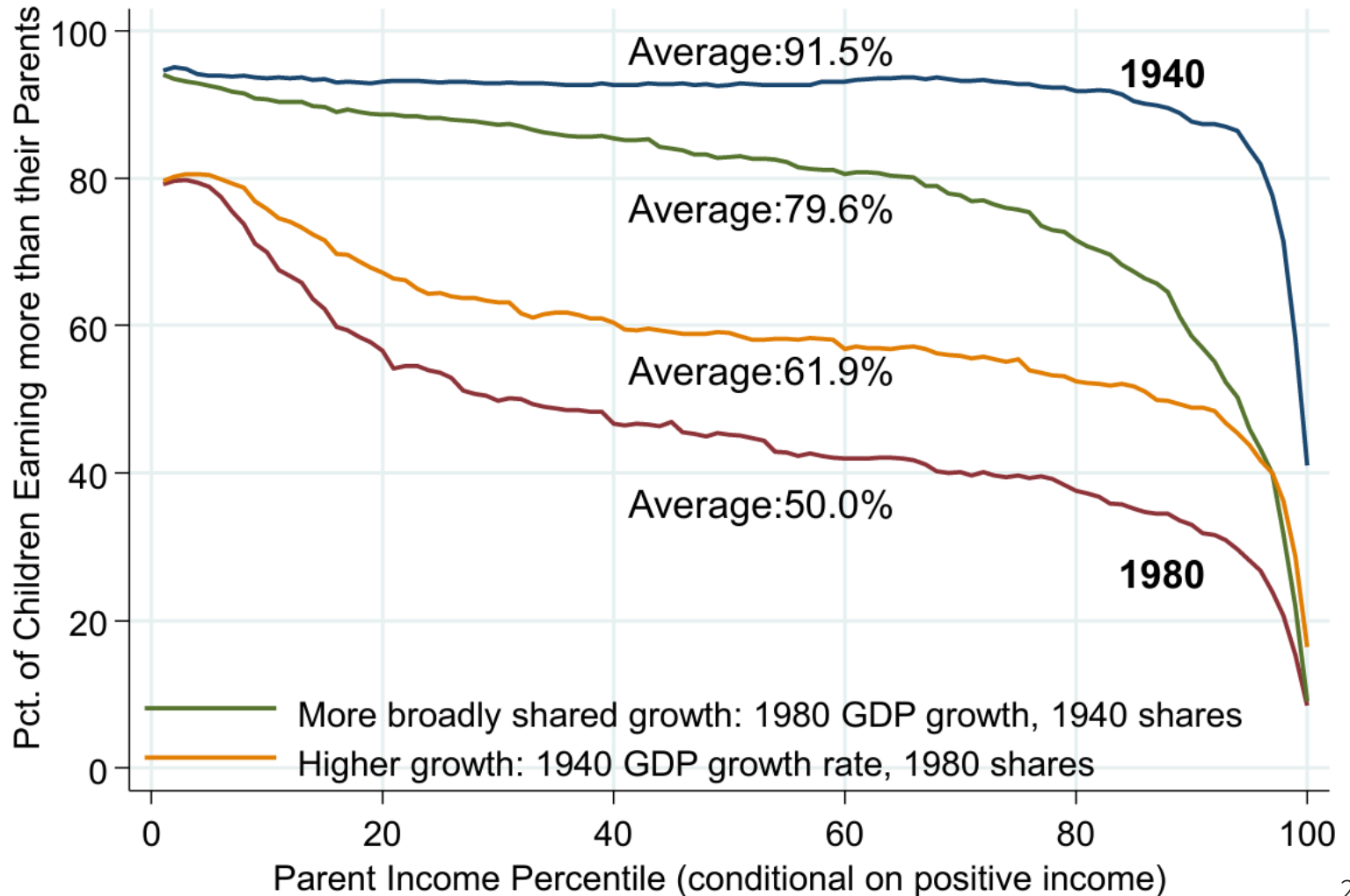
Hypothetical scenarios

Percent of Children Earning More than Their Parents: Hypothetical Scenarios



Hypothetical scenarios

Percent of Children Earning More than Their Parents: Hypothetical Scenarios



Restoring the American Dream

Main lesson: Restoring the American Dream of high rates of upward mobility will require **more broadly shared** economic growth

- Need policies that will increase incomes throughout the distribution

Two approaches of redistribution

(i) Redistribution

- Piketty and Saez: Reduction in top income taxes and erosion of unions and min wage have led working-class Americans to fall behind

(ii) Human Capital Investment

- Goldin and Katz: race between education and technology – need education to keep pace with technological change to increase wage rates

"Big data" and the American Dream

Measuring the American Dream

Central policy question:

*Why are children's chances of climbing the income ladder **falling** in America?*

*What can we do to **reverse** this trend?*

Difficult to answer using **historical data** and **macroeconomic trends**

Problem: We have too few data points to test all alternative explanations

- We never had the data to link parental income across generations

Measuring the American Dream

The **ideal data** to answer this question:

Individual by **year** observations across the American public including:

- family identifiers to link across generation
- residence information

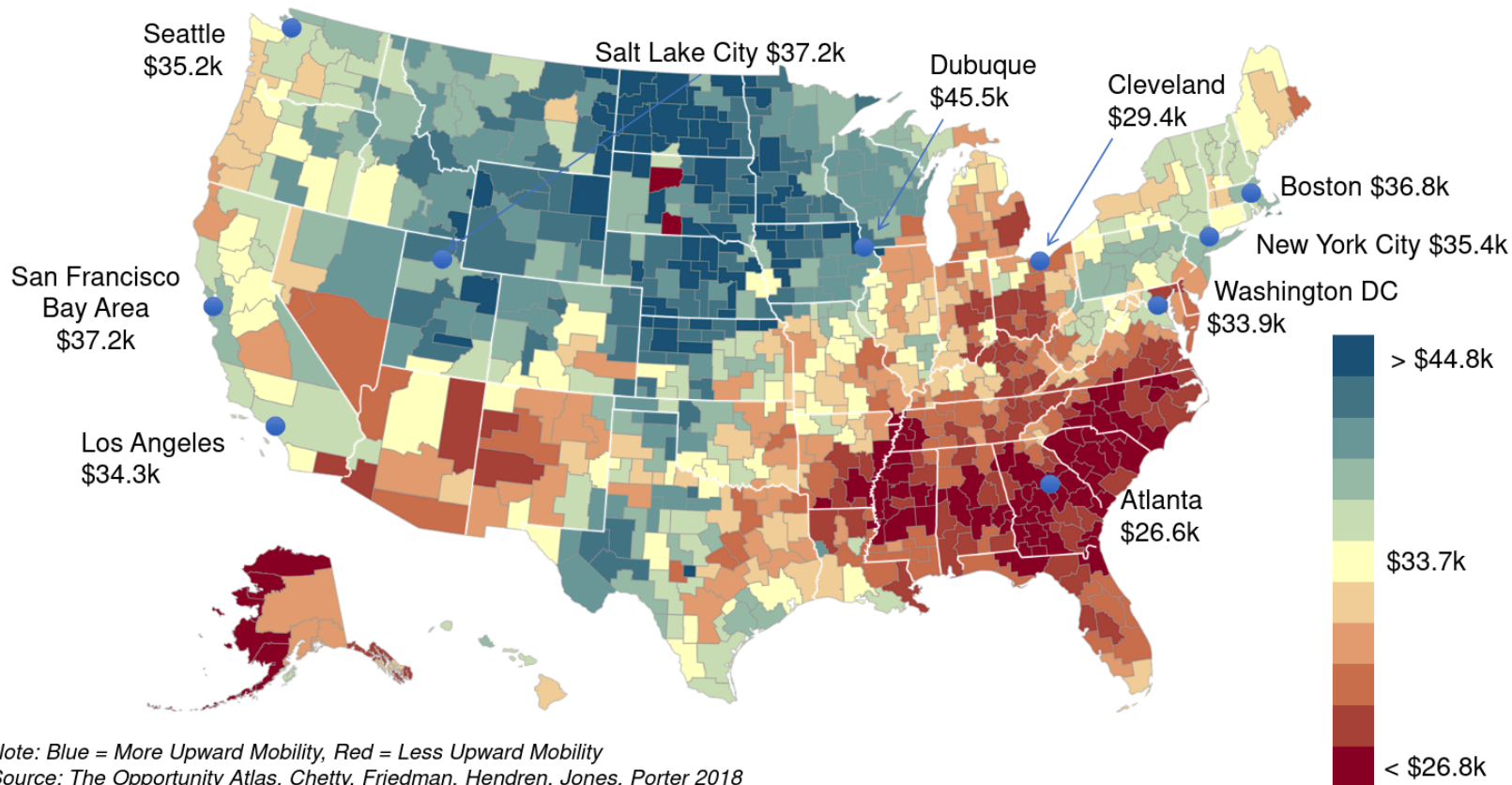
The universe of **tax records** fits the bill- linking generational income

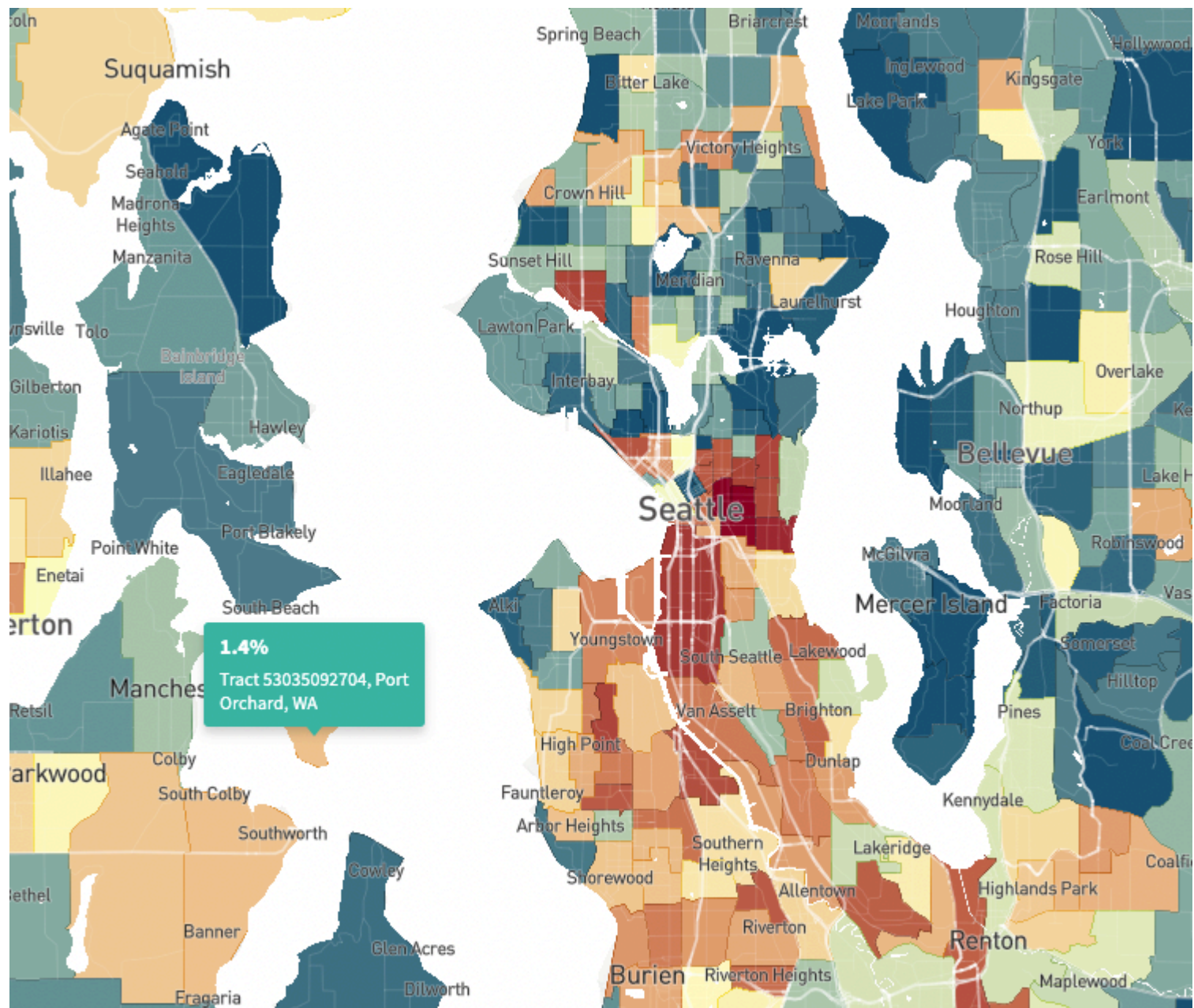
- Chetty et al. (2017)
- Tracking earnings over someone's life at the **individual level**

Allow researchers to observe **generational differences** in income

The Geography of Upward Mobility in the United States

Average Household Income for Children with Parents Earning \$27,000 (25th percentile)





Causal Effect of Neighborhoods

Causal Effect of Neighborhoods

The propensity for an individual to reach the American Dream seems to depend greatly on the location they were raised

Understanding why this **heterogeneity** exists may be key to understanding how to improve **social mobility**

*But is this relationship **causal**?*

How should we think about the explanations of these results?

There are two different explanations:

(i) Sorting: Similar people live near each other (educated people live near other educated people)

(ii) Causal impacts: places have a **causal effect** on upward mobility

Causal Effect of Neighborhoods

What is the **ideal experiment**?

Randomly assign children to neighborhoods. Compare adult outcome

- Can't do this? So what can we do?

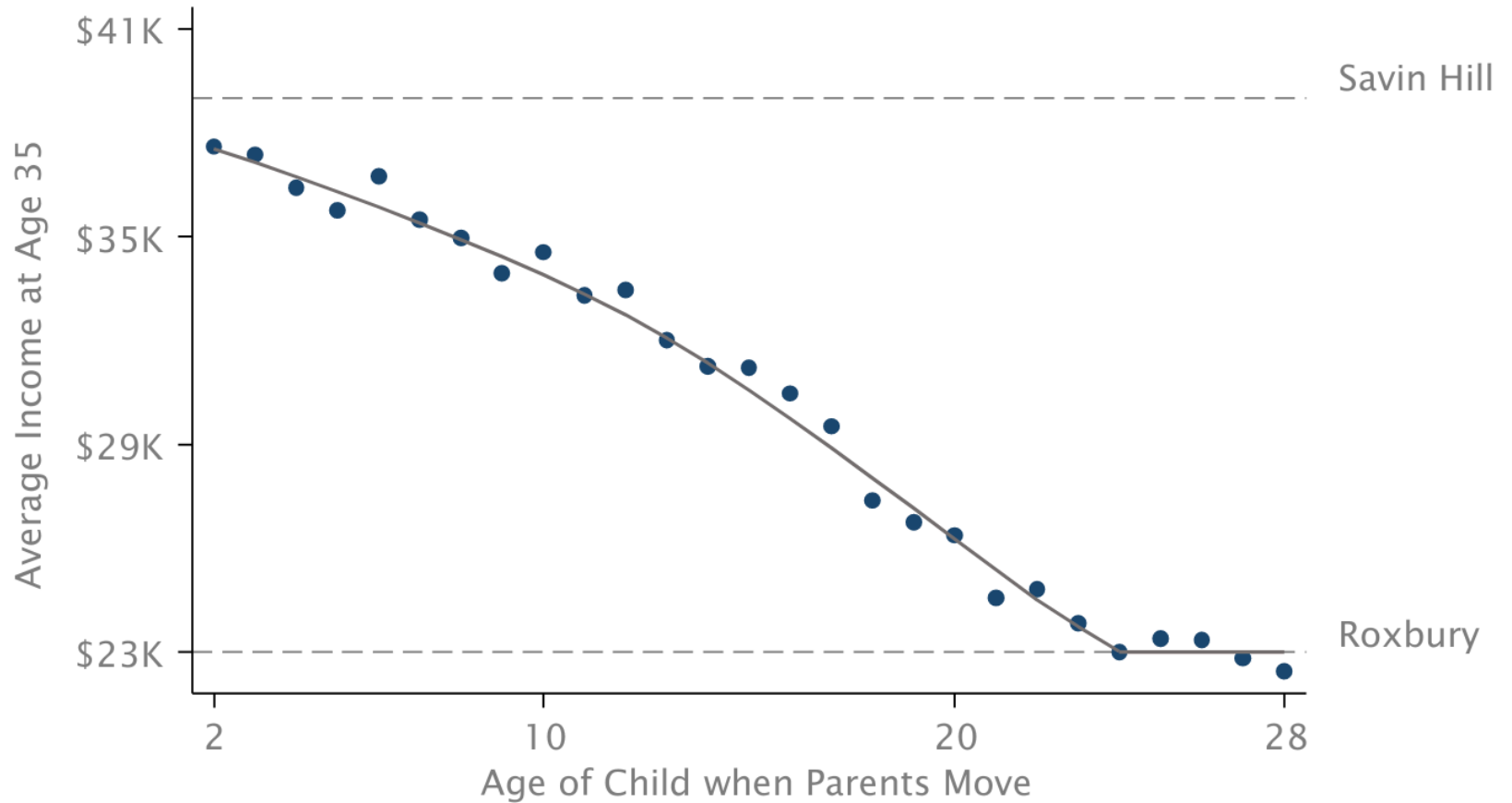
Chetty & Hendren (2018)

Quasi-experiment: Observe data on 7 million families that move

Key Idea: Exploit variation in *age of child* when family moves to identify causal impact of neighborhood environment

Income Gain from Moving to a Better Neighborhood

By Child's Age at Move



Assumptions

To identify causal impact of neighborhoods, you need to believe that **the timing of moves to better/worse areas are unrelated to other determinants of child's outcomes.**

This assumption might not hold for two reasons:

- (i)** Parents who move to good areas when their children are young might be different than those who move later
- (ii)** Moving may be unrelated to other factors (e.g change in parents' job) that impact children directly

Both of these concerns are addressed

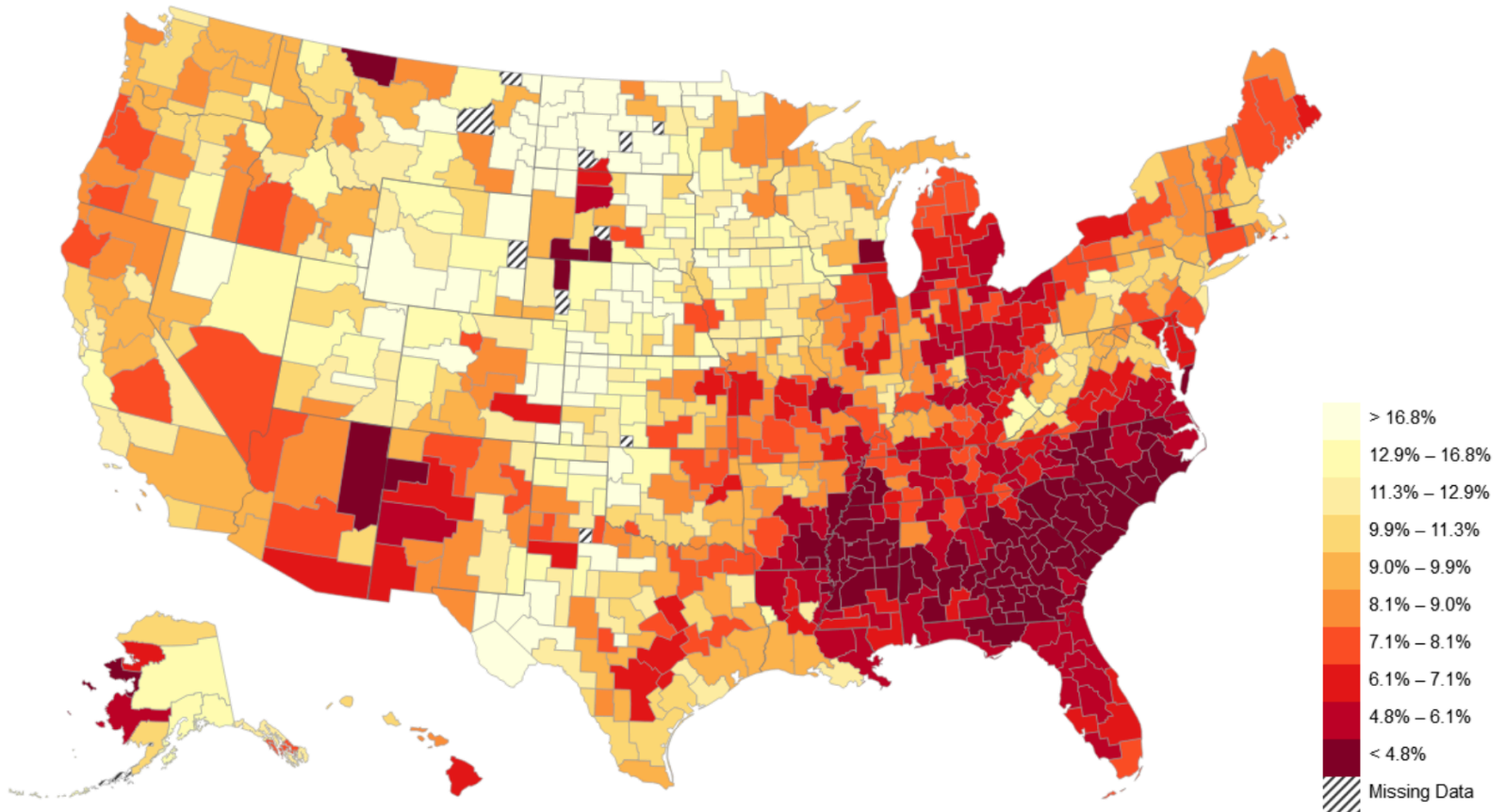
- Compare across siblings

Geography of Upward Mobility

A Familiar Map

The Geography of Upward Mobility in America

Children's Chances of Reaching Top 20% of Income Distribution Given Parents in Bottom 20%

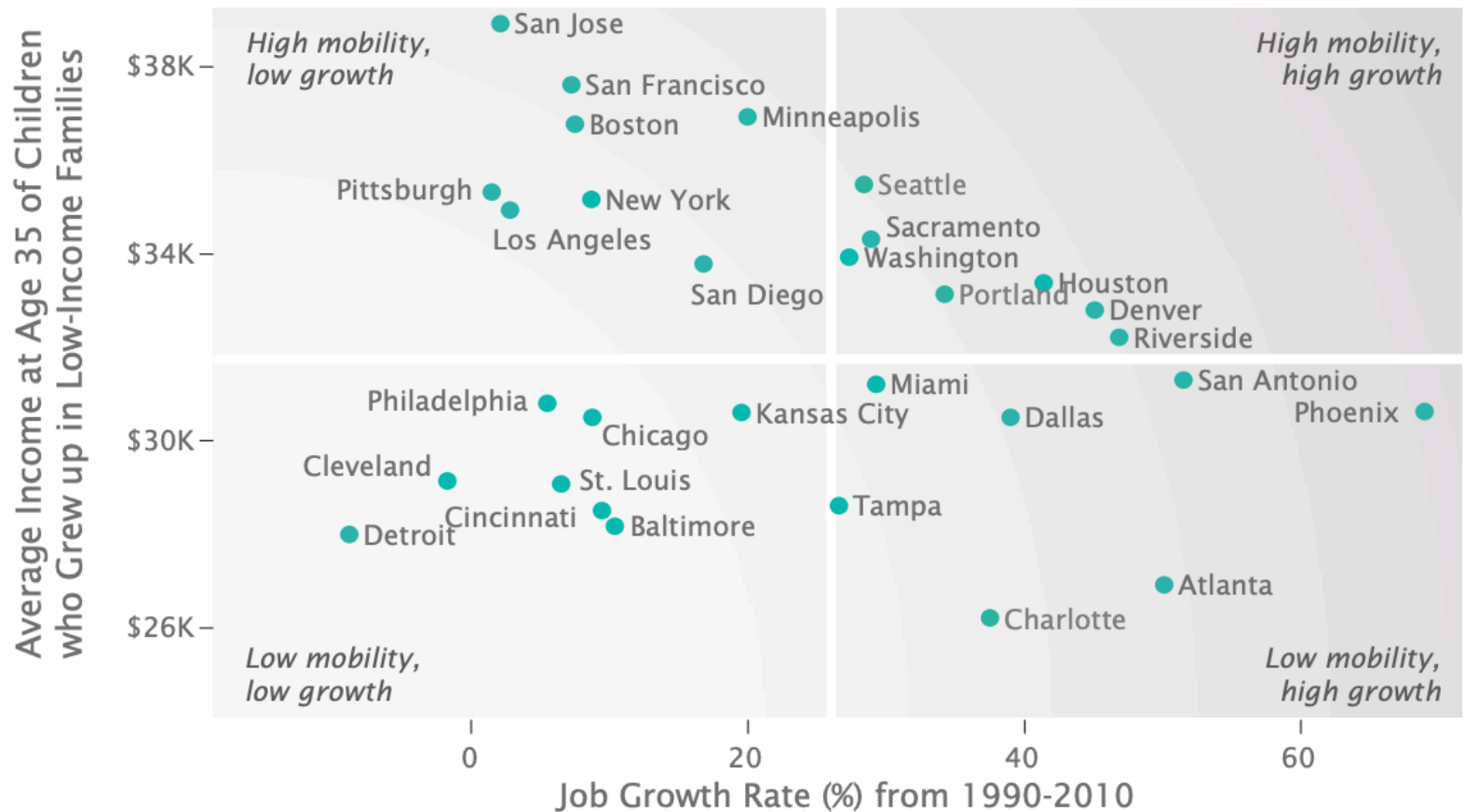


Last Q

OK, so: **neighborhoods matter for upward mobility**. Next question: *why does this vary across cities/places?*

- To answer this question, it would be good to start by comparing places with high upward mobility to low upward mobility
 - Do places with higher mobility have better schools, jobs, institutions, something else? All?

Correlations



Correlations

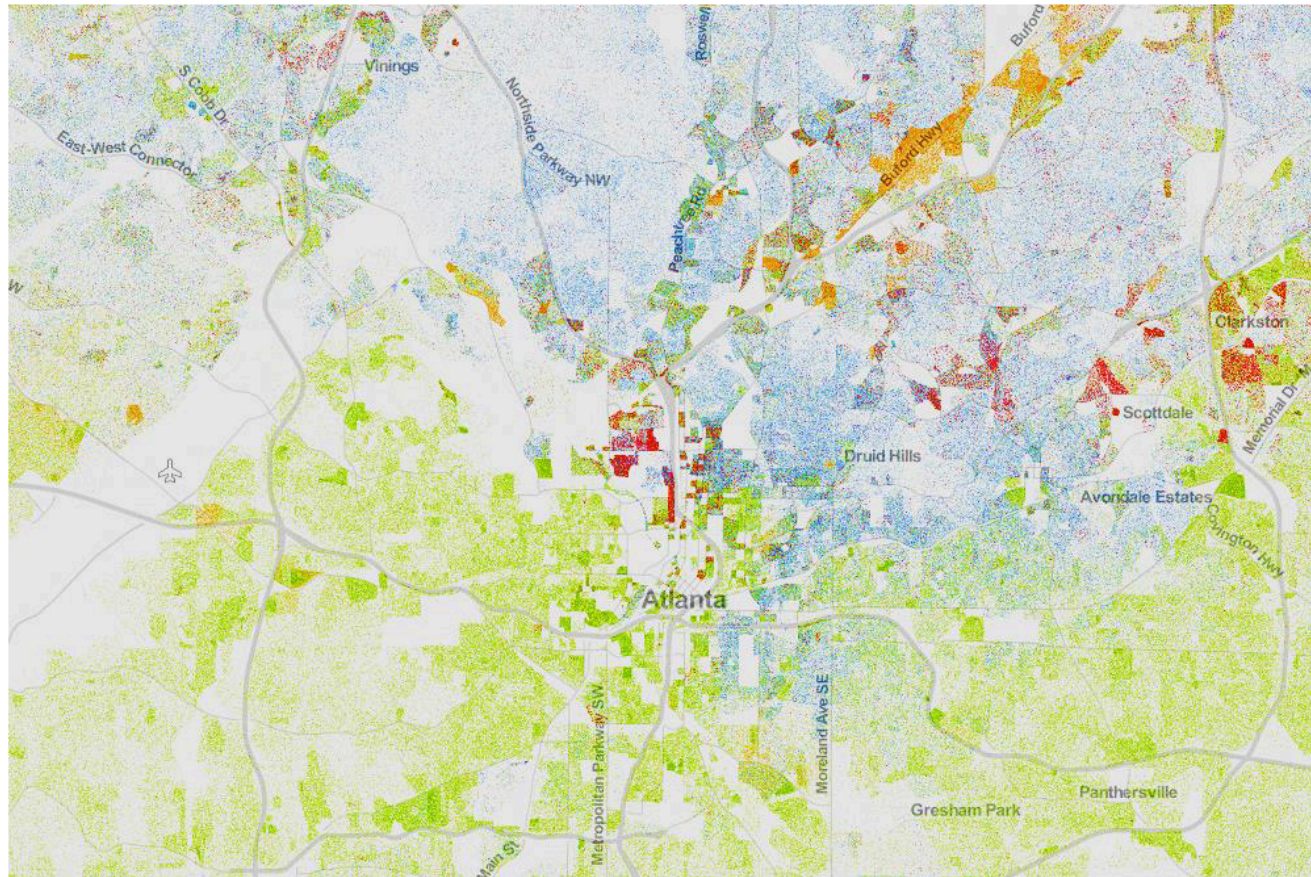
5 strongest correlates of upward mobility are:

(i) Segregation: Greater racial and income segregation associated with lower levels of mobility

Segregation

Racial Segregation in Atlanta

Whites (blue), Blacks (green), Asians (red), Hispanics (orange)



Correlations

5 strongest correlates of upward mobility are:

- (i) **Segregation:** Greater racial and income segregation associated with lower levels of mobility
- (ii) **Income Inequality:** Places with a smaller middle class have less mobility
- (iii) **School Quality:** Higher expenditure, smaller classes, higher test scores
⇒ more mobility
- (iv) **Family Structure:** Areas with more single parents have much lower mobility
 - Strong correlation even for kids whose *own* parents are married
- (v) **Social Capital:** "Takes a village to raise a child"

Policies to improve upward mobility

In general, there are two policy approaches to increasing upward mobility

(i) Moving to Opportunity (MTO): Provides affordable housing in high-opportunity areas

(ii) Place-Based Investments: Increases upward mobility in low-opportunity areas

Q: Can you think of the pros and cons of each approach?

MTO Experiment:

Implemented from 1994-1998 in 5 locations: Baltimore, Boston, Chicago, LA, NY

- 4,600 families randomly assigned to one of three groups:

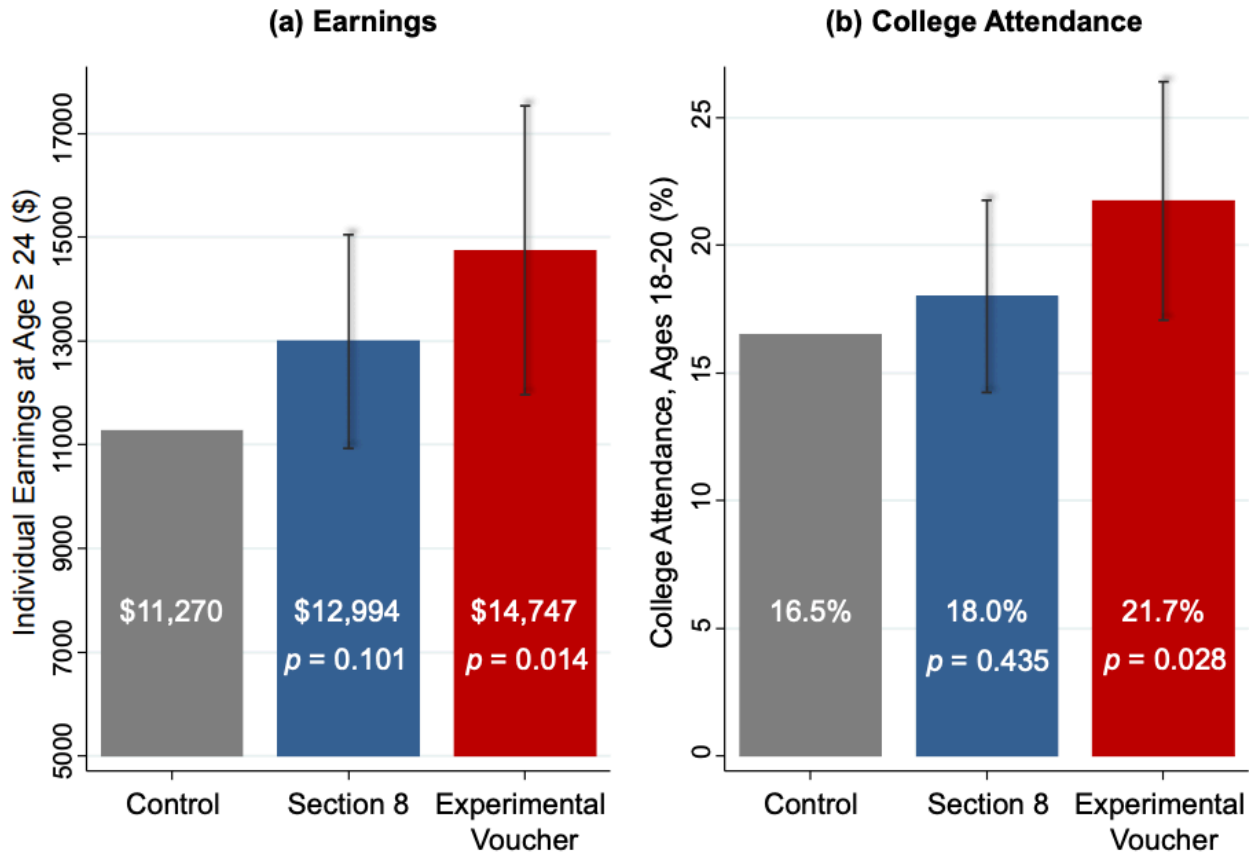
(i) Experimental: Offered housing vouchers restricted to low-poverty census tracts

(ii) Section 8: Offered conventional housing vouchers, no restrictions

(iii) Control: not offered a voucher, stayed in public housing

Results

Impacts of MTO on Children Below Age 13 at Random Assignment



Implications

Housing vouchers can be effective -- but should be target carefully

- Vouchers should be explicitly designed to help families move to affordable, high-opportunity areas
 - In MTO experiment, unrestricted vouchers produced **smaller** gains even though families could have made same moves
 - More generally, low-income families rarely use cash transfers to move to better neighborhoods
 - 80% of 2.1 Section 8 vouchers are currently used in high-poverty, low-opportunity neighborhoods

MTO Issues

What are the concerns with experiments like MTO?

(i) Costs: how costly would it be to scale this up?

(ii) Negative spillovers: does integration hurt the wealthy (which would dampen the effect of the spillovers)?

(iii) Limits to scaling due to sorting. If you move all low-income people into a high-income neighborhood, it is no longer a high-income neighborhood.

Place-Based Approaches

A **place-based approach** would be something like:

- (i) Investment in schools in high-poverty areas
- (ii) Investment in infrastructure in high-poverty areas

Q: What is the main problem with place-based approaches?

- Displacement! Locational eq $\implies$ more people sort into a neighborhood if it has higher school quality
- This drives up prices and the residents that it intended to help were ultimately harmed

Review

This course: In Review

We have finished all of the material in the class!

Thank you for your time and energy. A quick recap:

1) Big, philosophical questions

- What is a city? Why do they exist?
- What makes some cities grow and others shrink?

2) Tools!

- Locational equilibrium (x2)
- Bid-rent curves
- Two labor market models

This course: In Review

3) Applications:

- Min wage & rent control
- Land use & housing policy

My **big picture** takeaways:

- Location matters! The geographic level at which we implement policy matters, too
- Differences in structure of labor/housing market can lead to vastly different outcomes from the same policy
- Public policy questions are often more nuanced than they appear on a surface level

Thank you for a great class :)